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**Q 1.** *Use integration, the Direct Comparison Test, or the Limit Comparison Test to test the integrals for convergence.*

*a)* $\displaystyle\int_{-1}^{1} \frac{\mathrm{d}\theta}{\theta^2 - 2\theta}$



*b)* $\displaystyle\int_{0}^{2} \frac{\mathrm{d}x}{1 - x^2}$

*c)* $\displaystyle\int_2^{\infty} \frac{1}{\ln x}\,\mathrm{d}x$



*d)* $\displaystyle\int_{\pi}^{\infty} \frac{1+\sin x}{x^2}\,\mathrm{d}x$

**Q 2.** *Find the values of $p$ for which each integral converges:*

**a)** $\displaystyle\int_1^2 \frac{\mathrm{d}x}{x(\ln x)^p}$



**b)** $\displaystyle\int_2^\infty \frac{\mathrm{d}x}{x(\ln x)^p}$

**Q 3.** *Which of the sequences converge, and which diverge? Find the limit of each convergent sequence.*

**a)** $a_n = \dfrac{n + (-1)^n}{n}$

**b)** $a_n = \dfrac{1 - 5n^4}{n^4 + 8n^3}$



**c)** $a_n = \dfrac{n}{2^n}$

**d)** $a_n = \dfrac{n!}{10^{6n}}$

**Q 4.** *Use the nth-term test for divergence to show that the series is divergent, or state that the test is inconclusive:*

**a)** $\displaystyle\sum_{n=1}^{\infty} \frac{n(n+1)}{(n+2)(n+3)}$

**b)** $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^2+3}$



**Q 5.** *Determine whether the series converges or diverges. Find its sum if it converges:*

**a)** $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$

**b)** $\sum_{n=1}^{\infty}(\sqrt{n+4}-\sqrt{n+3})$

**Q 6.** *Which of the following series converge, and which diverge? Give reasons for your answers. If a series converges, find its sum.*



**a)** $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{3}{2^n}$

**b)** $\displaystyle\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$

**c)** $\displaystyle\sum_{n=0}^{\infty} \frac{\cos(n\pi)}{5^n}$